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Public Health

Unilateral Visual Impairment and Increased Mortality: The UK Biobank Cohort Study

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Received: January 29, 2025
Accepted: June 10, 2025
Published: July 23, 2025

Keywords: UK Biobank; cohort study; visual impairment (VI); mortality

Citation: Fu C, Chen J, Gu Q, Li Y, Li X, Zhang X, Chen W. Unilateral visual impairment and increased mortality: The UK Biobank cohort study. Transl Vis Sci Technol. 2025;14(7):17, https://doi.org/10.1167/tvst.14.7.17

Purpose: The purpose of this study was to investigate the association between the full spectrum of visual acuity categories and risks of all-cause, cardiovascular disease (CVD)-related, cancer-related mortality in the UK Biobank study.

Methods: A total of 131,468 participants (aged 40–79 years) with visual acuity measurement from the UK Biobank prospective cohort were included. Visual acuities of both eyes were used to assign each participant into one of the following groups: bilateral normal, unilateral near normal, bilateral near normal, unilateral visual impairment (VI), mild VI, and low vision. Cox proportional hazards multivariable regression models were used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs) for the risks of all-cause, CVD-related, and cancer-related mortality.

Results: Over a median follow-up of 13.49 years, a total of 9372 deaths were recorded, including 4682 cancer-related deaths and 1926 CVD-related deaths. Compared with the bilateral normal group, participants with decreased visual acuity had increased risks of all-cause and CVD-related mortality with dose-response-type gradients, but no elevated risk of cancer-related mortality was observed. Decreased unilateral visual acuity but not bilateral near normal vision was associated with increased risks for all-cause and CVD-related mortality. For all-cause and CVD-related mortality, the adjusted HRs were 1.17 (95% CI = 1.09–1.25) and 1.24 (95% CI = 1.07–1.44) for the unilateral near normal group, and 1.20 (95% CI = 1.14–1.27) and 1.26 (95% CI = 1.12–1.42) for the unilateral VI group.

Conclusions: Decreased unilateral visual acuity was associated with increased all-cause and CVD-related mortality risks among participants without VI in the UK Biobank.

Translational Relevance: Achieving bilateral vision balance may contribute to reducing the mortality risk of individuals with decreased unilateral visual acuity.

Introduction

Visual impairment (VI), encompassing conditions from mild visual difficulties to complete blindness, has a profound impact on individuals, their families, and society.[1,2] VI is associated with increased risks of cognitive impairment, dementia, depression, psychosis, and Parkinson's disease, and with reduced quality of life and loss of independence, posing a substantial public health challenge.[3–12] Studies across diverse populations and countries investigated the relationship between VI and increased mortality risks.[13–15] Some studies also reported the association between VI and cause-specific mortality risks, including cardiovascular disease (CVD)-related, cancer-related, and injury-related mortality risks.[16–18] However, most existing studies focused primarily on individuals with severely

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impaired visual acuity, leaving a gap in understanding mortality risks for those with mildly reduced or near-normal visual acuity. The gap may be partly attributable to the widespread use of the World Health Organization (WHO) VI classification, which defines VI as visual acuity in the better eye worse than 6/12 (Snellen equivalent).[1]

Individuals with mildly reduced visual acuity faced elevated risks of lacking educational qualifications, possessing higher deprivation scores, being unemployed, holding lower-status jobs, living alone, and experiencing mental health issues.[19,20] These challenges contribute significantly to societal burdens, especially considering the large proportion of individuals with mildly reduced visual acuity worldwide. A recent study involving 581,526 Korean adults found that individuals with mildly reduced visual acuity faced a 17% increased risk of all-cause mortality among young to middle-aged population (average age = 39.7 years), but not in the overall population or among older adults.[17] Thus, it underscored the need for comprehensive studies that consider the entire spectrum of visual acuity categories to better understand and address the broader impacts of visual acuity on mortality.

On the other hand, in previous studies exploring the association between VI and mortality risk, VI was defined using various measures: presenting visual acuity of the better eye,[17] binocular presenting visual acuity, or best-corrected visual acuity of the better or worse eye.[13] For each individual, monocular or binocular visual acuity was used to identify the vision level, which, however, could not reflect the combined visual acuities of both eyes. Individuals with visual acuity above the VI threshold in one eye and below the threshold in the other eye were overlooked and classified into the reference group along with those who had good visual acuities in both eyes.

Fortunately, the UK Biobank Eye and Vision Consortium offers a comprehensive classification system that covers the full spectrum of visual acuity categories by assessing the visual acuities of both eyes. This system classified visual acuity levels above the WHO VI threshold (equal to or better than 6/12) into normal (6/6 to 6/9.5) and near-normal vision (worse than 6/9.5 to 6/12), encompassing a spectrum from bilateral normal vision to blindness. It further subdivides individuals within the WHO-defined non-VI group into four categories: bilateral normal vision (normal vision for both eyes), unilateral near-normal vision (normal vision for one eye and near-normal vision for the other), bilateral near-normal vision (near-normal vision for both eyes), and unilateral VI (above the WHO VI threshold for one eye and below for the other).[19] Leveraging the UK Biobank dataset, the largest prospective adult cohort with visual acuity measurements and over a decade of follow-up,[21] our study aimed to investigate the associations between the full spectrum of visual acuity categories and mortality risks, including all-cause, CVD-related, and cancer-related mortality risks. Additionally, we explored these associations across age subgroups (above and below 60 years) and by sex stratification. We hypothesized that, even among individuals classified as non-VI according to WHO criteria, variations in visual acuity levels would be associated with mortality risk, and that these associations would differ by age and sex.

Methods

Study Design and Population

The datasets used in this study were obtained from the UK Biobank (Application: 129237). The UK Biobank is a large, population-based prospective cohort, in which more than 0.5 million participants aged 37 to 73 years attended one of the 22 assessment centers across England, Scotland, and Wales between 2006 and 2010 at baseline examinations (5.5% response rate).[22,23] Between 2009 and 2010, approximately 117,252 participants underwent the initial assessment visit including eye measurements. An additional 16,016 participants, who had not received eye examinations at their initial visits, underwent eye assessments during the first repeat assessment visit between 2012 and 2013.[24] We defined the visit with the first available visual acuity examination as the "baseline", either between 2009 and 2010 or 2012 and 2013.

The UK Biobank study was granted ethical approval from the National Health Service (NHS) North West Centre for Research Ethics, adhering to the principles outlined in the Declaration of Helsinki (Ref: 21/NW/0157). This work used data provided by patients and collected by the NHS as part of their care and support. All participants gave written informed consent and signed the informed consent to be linked to national electronic health-related datasets.

Classification of Visual Acuity Categories

Habitual distance visual acuity in each eye (i.e. tested with the individual wearing any prescribed, presently used optical correction) was measured to assess real-life visual acuity rather than best-corrected

visual acuity (Field ID 5208 and 5201, measurement details in Resource 100250). Briefly, visual acuity was measured using a traditional logarithm of the minimum angle of resolution (LogMAR) chart on a computer screen under standard illumination.[25]

Each individual was assigned to one of six mutually exclusive visual acuity categories based on the LogMAR visual acuity of both eyes. According to the WHO definition, VI was defined as visual acuity worse than 6/12 (> 0.3 LogMAR) in the better-seeing eye. Above the WHO VI threshold, we defined normal vision as visual acuity between 0 and 0.2 (6/6 to 6/9.5, Snellen equivalent), and near-normal vision as visual acuity between 0.21 and 0.3 (worse than 6/9.5 to 6/12, Snellen equivalent). Considering the combined visual acuities of both eyes, individuals not meeting the WHO criteria for VI (i.e. the WHO non-VI group) were subdivided into four groups: (1) bilateral normal vision (visual acuity of 0–0.2 in both eyes); (2) unilateral near-normal vision (0–0.2 in one eye and 0.21–0.3 in the other eye); (3) bilateral near-normal vision (0.21–0.3 in both eyes); and (4) unilateral VI (0–0.3 in one eye and 0.31 or worse in the other eye). In addition, individuals who met the WHO definition of VI were classified as (5) mild VI (visual acuity of 0.31–0.49 in the better eye), and (6) low vision (visual acuity $\geq$ 0.5 in the better eye). The low vision group included participants from the WHO moderate VI (n = 966), WHO severe VI (n = 71), and WHO blindness categories (n = 4), due to the small sample sizes within each individual group. Detailed definitions and the distribution of WHO VI classification and six visual acuity categories are provided in Supplementary Table S1.

Mortality and Follow-Up

The date of death (Field ID 40000) was obtained from NHS Digital for participants in England and Wales and from the NHS Central Register for those in Scotland, with a data update period of every 3 or 6 months. It was used to determine the survival status and follow-up endpoint. Participants not matched with death certificates were considered alive. Participant follow-up duration started at the date of baseline and ended either at the date of death or October 13, 2023. The underlying cause of death (Field ID 40001) was used to determine the specific cause of death according to the International Statistical Classification of Diseases and Related Health Problems, 10th revision (ICD-10). CVD-related mortality was identified using ICD-10 codes I00 to I99, whereas cancer-related mortality was determined using ICD-10 codes C00 to D48.

Covariates Ascertainment

In this study, demographic characteristics (age, sex, ethnicity, income, and education), lifestyle factors (smoking and drinking status), medical history (diabetes mellitus, hypertension, stroke, and heart attack histories), and body mass index (BMI) were collected. Sex was acquired from the central registry at recruitment (Field ID 31). The baseline information was collected during the baseline visit, including age (Field ID 21003), income (Field ID 738), ethnicity (Field ID 21000), education (Field ID 6138), smoking and drinking status (Field ID 20116 and 20117), medical history (hypertension, stroke, and heart attack histories from Field ID 61500, and diabetes mellitus from Field ID 2443) and BMI (kg/m^2, Field ID 61500). BMI was imputed with mean value (missing percent = 0.50%) and categorized into four groups according to the WHO BMI classification.[21]

Statistical Analysis

Participant characteristics were summarized utilizing descriptive statistics. Continuous variables were presented as mean $\pm$ standard deviation or median (interquartile range [IQR]), depending on the normality of data distribution, which was evaluated using Q-Q plots. Categorical variables were presented as frequencies and percentages (n, %). The standardized mean difference was used to compare variables between visual acuity categories, with a standardized mean difference exceeding 0.1 indicating a significant difference in variables.

Cox proportional hazards regression was used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs) for all-cause, CVD-related, and cancer-related mortality in different visual acuity categories, with the bilateral normal group as the reference. The assumption of proportional hazards was evaluated by statistical tests based on weighted Schoenfeld residuals,[26] and the stratification method was used to deal with the violation of variables sex and age. Model 1 was stratified with sex and age group (age $\geq$ 60 or < 60 years). Model 2 was adjusted for age and ethnicity based on model 1. Model 3 was further adjusted for income, education, smoking and drinking status, histories of diabetes mellitus, hypertension, stroke, and heart attack, and BMI category based on model 2. Based on our hypothesis and the violation of the proportional hazard assumption, subgroup analyses were performed with stratification by sex and age.

Several sensitivity analyses were carried out to assess the robustness of our results. First, participants with cancer or CVD at baseline were excluded

when assessing the associations between visual acuity categories and specific-cause mortality. Second, we excluded participants who died within the first 2 or 4 years of follow-up to avoid the potential risk of reverse causation. Third, Fine and Gray subdistribution hazard models were used to examine the associations between visual acuity categories and CVD-related and cancer-related mortality with the death of other causes considered as competing events.[27]

Statistical analyses were performed utilizing R software (version 4.3.0, https://www.R-project.org/). All statistical tests were 2-sided, and the type I error rate for statistical significance was set at 0.05.

Results

Baseline Characteristics

A total of 131,468 participants were included in the final analysis, representing 26.18% of the original UK Biobank cohort of 502,165 individuals. The details of the participants' exclusion process are presented in Figure 1. The mean age of these participants was 57.42 ± 8.17 years, with 45.95% being male participants. The distribution of visual acuity categories among participants was as follows: 75.28% had bilateral normal vision, 7.76% had unilateral near-normal vision, 0.87% had bilateral near-normal vision, 13.18% had unilateral VI, 2.12% had mild VI, and 0.79% had low vision (definition, distribution, and the comparison with WHO VI classification provided in Supplementary Table S1). The Table presents an overview of participant characteristics, stratified by visual acuity categories. There were significant differences in age, ethnicity, income, and highest educational attainment between visual acuity categories.

Association of Visual Acuity Categories With All-Cause Mortality

Over a median follow-up period of 13.49 years, a total of 9372 deaths were recorded. Survival curves in Figure 2A show that participants with decreased visual acuity had lower long-term survival rates compared with those with bilateral normal vision. Figure 2B shows the association between decreased visual acuity and increased all-cause mortality risk, and the strength of the association increased from unilateral near normal to low vision (P for trend < 0.001) in model 3 (source data and results of model 1 and model 2 in Supplementary Table S2). Specifically, the adjusted HRs were 1.17 (95% CI = 1.09–1.25) for the unilateral near-normal vision group and 1.20 (95%

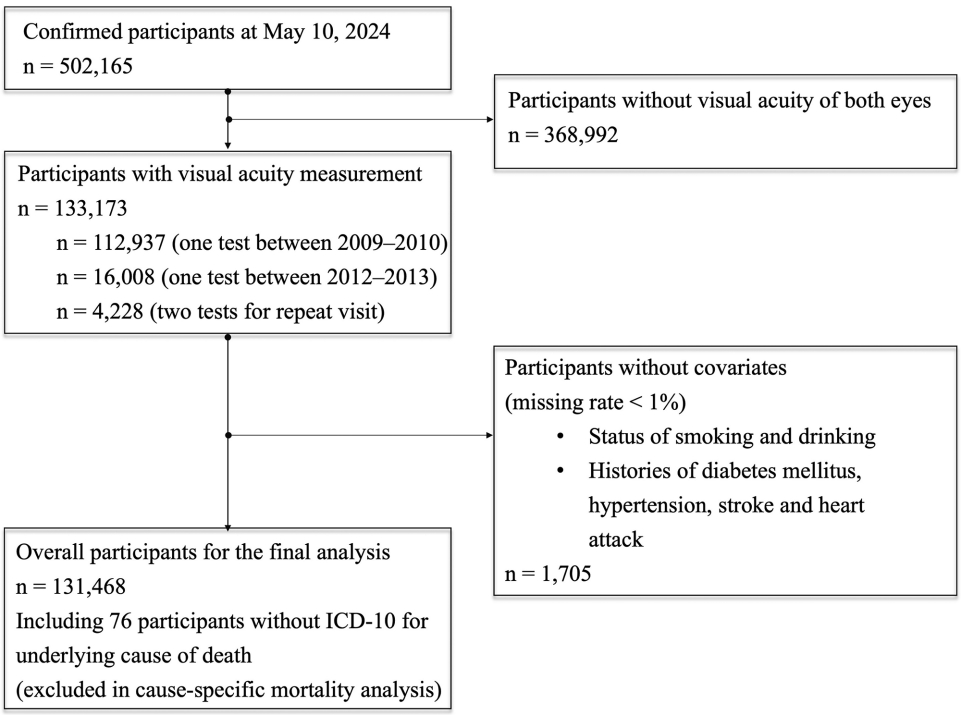


Figure 1. Flowchart of participants' exclusion process. ICD-10, International Statistical Classification of Diseases and Related Health Problems, 10th revision.

Table. Characteristics of Participants Stratified by Visual Acuity Categories

	Overall $n = 131,468$	Bilateral Normal $n = 98,974$	Unilateral Near Normal $n = 10,203$	Bilateral Near Normal $n = 1,141$	Unilateral VI $n = 17,326$	Mild VI $n = 2,783$	Low Vision $n = 1,041$	*P* Value	SMD
Sex, male, *n* (%)	60,416 (45.95)	45,576 (46.05)	4,556 (44.65)	496 (43.47)	8,020 (46.29)	1,311 (47.11)	457 (43.90)	0.015	0.036
Age, y, mean ± SD	57.42 (8.17)	56.84 (8.28)	59.35 (7.52)	60.35 (7.08)	59.02 (7.59)	59.66 (7.40)	57.94 (8.07)	<0.001	0.200
Ethnicity, *n* (%)									
White	115,338 (87.73)	87,319 (88.22)	8,894 (87.17)	975 (85.45)	14,972 (86.41)	2,332 (83.79)	846 (81.27)	<0.001	0.113
Mixed	6,711 (5.10)	4,683 (4.73)	557 (5.46)	71 (6.22)	1,085 (6.26)	231 (8.30)	84 (8.07)		
Asian or Asian British	5,814 (4.42)	4,422 (4.47)	418 (4.10)	52 (4.56)	740 (4.27)	132 (4.74)	50 (4.80)		
Black or Black British	1,241 (0.94)	904 (0.91)	116 (1.14)	15 (1.31)	167 (0.96)	23 (0.83)	16 (1.54)		
Chinese	548 (0.42)	373 (0.38)	57 (0.56)	5 (0.44)	91 (0.53)	15 (0.54)	7 (0.67)		
Other	1,816 (1.38)	1,273 (1.29)	161 (1.58)	23 (2.02)	271 (1.56)	50 (1.80)	38 (3.65)		
Income, > £18,000 £, *n* (%)	89,138 (67.80)	69,179 (69.90)	6,323 (61.97)	665 (58.28)	10,931 (63.09)	1,481 (53.22)	559 (53.70)	<0.001	0.113
Highest educational attainment, *n* (%)									
No qualifications	18,481 (14.06)	12,430 (12.56)	1,857 (18.20)	251 (22.00)	3,063 (17.68)	661 (23.75)	219 (21.04)	<0.001	0.113
O levels	34,129 (25.96)	25,832 (26.10)	2,642 (25.89)	273 (23.93)	4,410 (25.45)	704 (25.30)	268 (25.74)		
A levels	15,232 (11.59)	11,798 (11.92)	1,059 (10.38)	111 (9.73)	1,906 (11.00)	272 (9.77)	86 (8.26)		
University/others	62,342 (47.42)	48,087 (48.59)	4,507 (44.17)	484 (42.42)	7,731 (44.62)	1,085 (38.99)	448 (43.04)		
Current smoker, *n* (%)	12,381 (9.42)	9,047 (9.14)	978 (9.59)	118 (10.34)	1,793 (10.35)	306 (11.00)	139 (13.35)	<0.001	0.054
Current drinker. *n* (%)	120,070 (91.33)	90,973 (91.92)	9,186 (90.03)	1,009 (88.43)	15,577 (89.91)	2,423 (87.06)	902 (86.65)	<0.001	0.079
History of diabetes mellitus, *n* (%)	7,738 (5.89)	5,375 (5.43)	705 (6.91)	96 (8.41)	1,254 (7.24)	219 (7.87)	89 (8.55)	<0.001	0.054
History of hypertension, *n* (%)	36,018 (27.40)	25,999 (26.27)	3,104 (30.42)	362 (31.73)	5,328 (30.75)	909 (32.66)	316 (30.36)	<0.001	0.053
History of stroke, *n* (%)	2,008 (1.53)	1,338 (1.35)	174 (1.71)	32 (2.80)	365 (2.11)	66 (2.37)	33 (3.17)	<0.001	0.057
History of heart attack, *n* (%)	2,889 (2.20)	1,962 (1.98)	259 (2.54)	29 (2.54)	531 (3.06)	81 (2.91)	27 (2.59)	<0.001	0.028
BMI categories, *n* (%)									
Normal	43,403 (33.01)	32,971 (33.31)	3,322 (32.56)	370 (32.43)	5,538 (31.96)	850 (30.54)	352 (33.81)	<0.001	0.049
Underweight	717 (0.55)	536 (0.54)	60 (0.59)	4 (0.35)	94 (0.54)	15 (0.54)	8 (0.77)		
Overweight	55,934 (42.55)	42,159 (42.60)	4,288 (42.03)	481 (42.16)	7,395 (42.68)	1,199 (43.08)	412 (39.58)		
Obesity	31,414 (23.89)	23,308 (23.55)	2,533 (24.83)	286 (25.07)	4,299 (24.81)	719 (25.84)	269 (25.84)		

BMI, body mass index; SD, standard deviation; SMD, standardized mean difference; VI, visual impairment.

Data were reported as mean ± SD for continuous variables and number (proportion) for categorical variables.

SMD exceeding 0.1 indicates a significant difference in variables, and *P* values were derived from one-way ANOVA tests for continuous variables and chi-square tests for categorical variables.

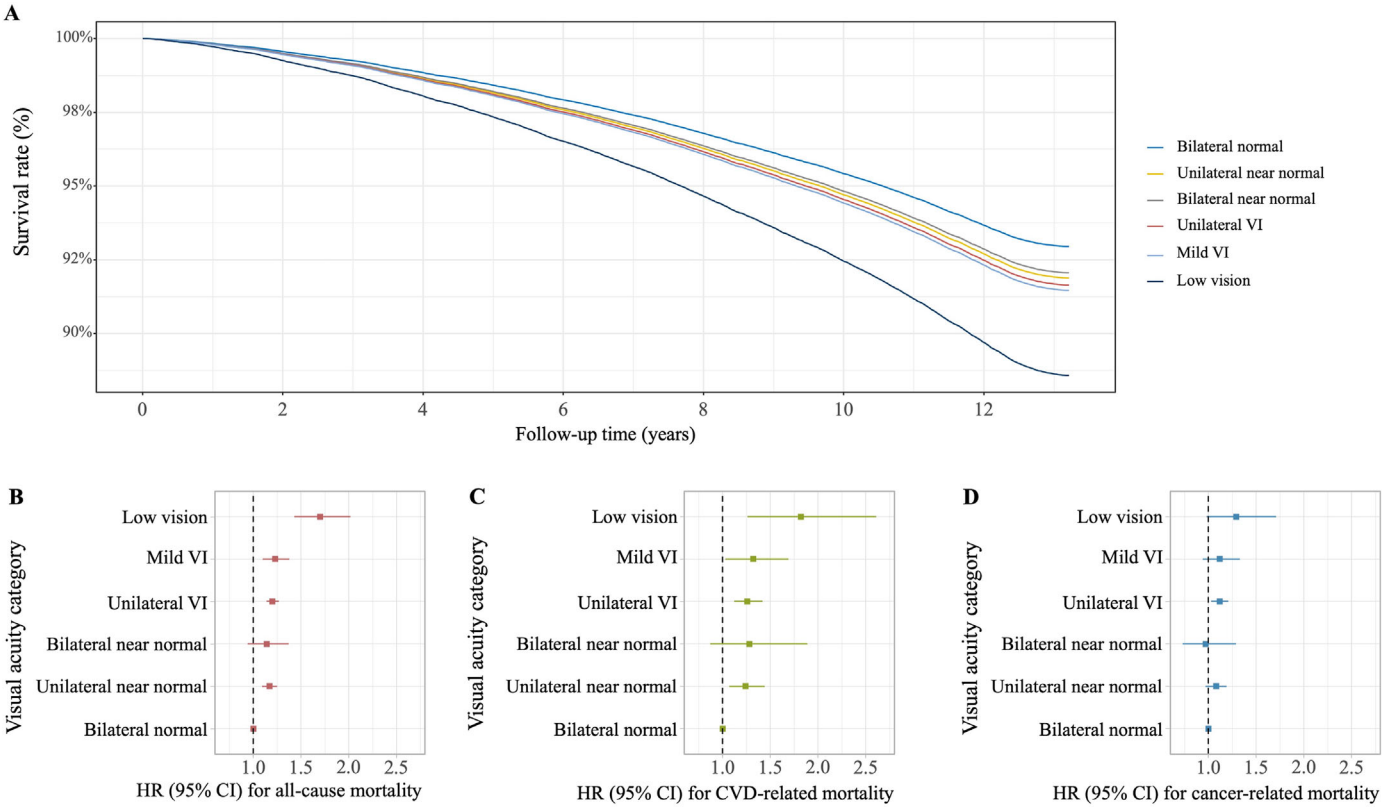


Figure 2. Associations of visual acuity categories with all-cause and cause-specific mortality among overall participants. (**A**) Adjusted survival curves of visual acuity categories. (**B**) Associations of visual acuity categories with all-cause mortality. (**C**) Associations of visual acuity categories with CVD-related mortality. (**D**) Associations of visual acuity categories with cancer-related mortality. Adjusted survival curves based on the Cox regression model stratified with sex and age group (age $\geq$ 60 or $<$ 60 years) and adjusted for age, ethnicity, income, education, smoking and drinking status, histories of diabetes mellitus, hypertension, stroke, and heart attack, and BMI categories. All analyses were stratified with sex and age group (age $\geq$ 60 or $<$ 60 years) and adjusted for age, ethnicity, income, education, smoking and drinking status, histories of diabetes mellitus, hypertension, stroke and heart attack, and BMI categories. Data are presented as HR $\pm$ 95% CI for Cox regression. Error bars represent the 95% CIs for each effect estimate. BMI, body mass index; CI, confidence intervals; CVD, cardiovascular disease; HR, hazard ratio, VI, visual impairment. Source data are provided in Supplementary Table S2.

CI = 1.14–1.27) for the unilateral VI group. Participants within the WHO VI group had increased risks with adjusted HRs of 1.23 (95% CI = 1.10–1.38) for the mild VI group and 1.70 (95% CI = 1.43–2.02) for the low vision group. However, the association between bilateral near-normal vision and all-cause mortality risk was not statistically significant in the fully adjusted model.

Association of Visual Acuity Categories With Cause-Specific Mortality

A total of 4682 cancer deaths (49.96% of all deaths) and 1926 CVD deaths (20.55% of all deaths) were recorded. Similar associations between decreased visual acuity and increased mortality risk were observed for CVD-related mortality as for all-cause mortality (Fig. 2C). Individuals with decreased visual acuity exhibited an increased risk of CVD-related mortality, except for the bilateral near-normal vision group, demonstrating a clear dose-response relationship (P for trend $<$ 0.001). For cancer-related mortality, the strength trend and associations with decreased visual acuity were observed in model 1 and model 2 (see Supplementary Table S2). However, the associations did not remain significant in the fully adjusted model (model 3), with one exception of association with unilateral VI, which remained significant after adjusting all potential confounders (Fig. 2D).

Stratified Analyses

In participants aged 60 years and older, similar associations between decreased visual acuity and increased risks of all-cause, CVD-related, and cancer-

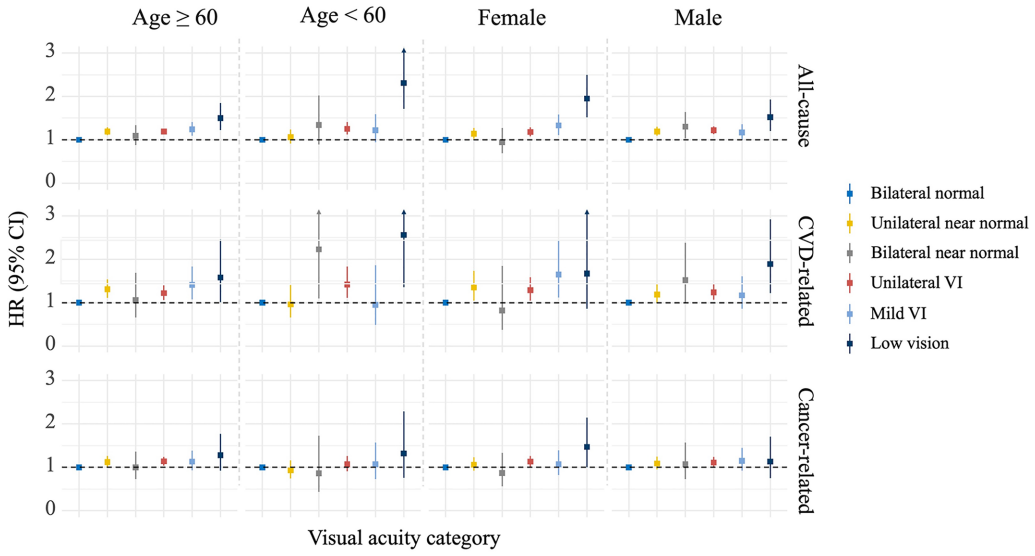


Figure 3. Associations of visual acuity categories with all-cause and cause-specific mortality among sex or age group stratified participants. All analyses were stratified with sex (age group stratification) or age group (sex stratification) and adjusted for age, ethnicity, income, education, smoking and drinking status, histories of diabetes mellitus, hypertension, stroke and heart attack, and BMI category. Data are presented as HR ± 95% CI for Cox regression. Error bars represent the 95% CIs for each effect estimate. *Arrows* indicate that the CIs are outside the range shown in the figure. Source data are provided in Supplementary Table S4. Results of model 1 and model 2 are provided in Supplementary Table S6.

related mortality were observed as those in the overall population. Among younger participants, unilateral VI and low vision were associated with an increased risk of all-cause mortality, whereas bilateral near normal, unilateral VI, and low vision were associated with an increased risk of CVD-related mortality. In female participants, the associations between decreased visual acuity and increased risks of all-

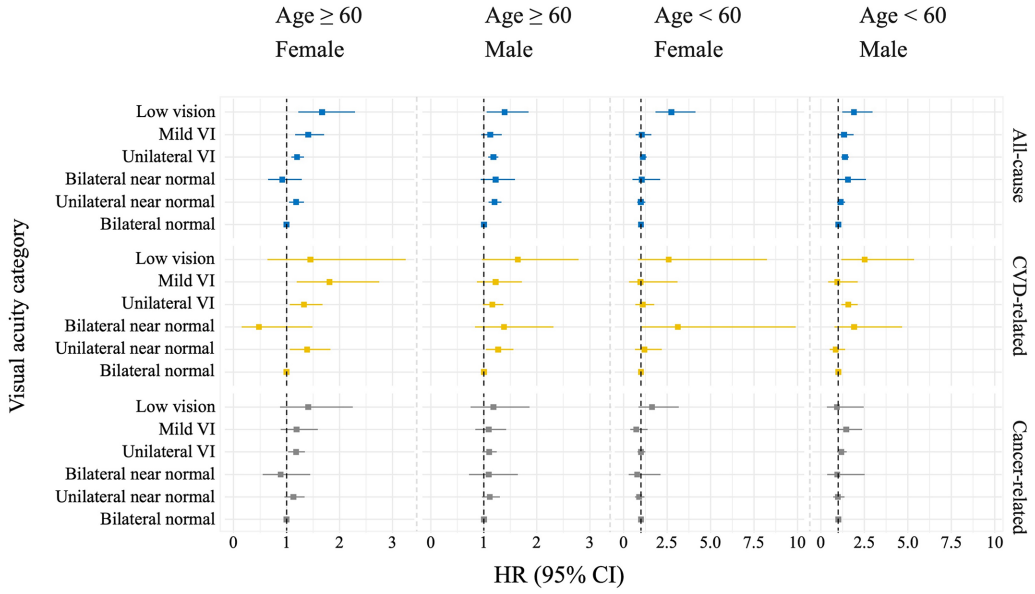


Figure 4. Associations of visual acuity categories with all-cause and cause-specific mortality among sex and age group stratified participants. All analyses were adjusted for age, ethnicity, income, education, smoking and drinking status, histories of diabetes mellitus, hypertension, stroke and heart attack, and BMI category. Data are presented as HR ± 95% CI for Cox regression. Error bars represent the 95% CIs for each effect estimate. Source data are provided in Supplementary Table S5. Results of model 1 and model 2 are provided in Supplementary Table S7.

cause, CVD-related, and cancer-related mortality were consistent with those observed in the overall population, except for the associations between low vision and risks of CVD-related and cancer-related mortality. In male participants, all levels of decreased visual acuity in the non-VI group and low vision were associated with an increased risk of all-cause mortality. Moreover, male participants with unilateral VI and low vision exhibited an increased risk of CVD-related mortality (Fig. 3, source data in Supplementary Table S4).

In subgroup analyses stratified by sex and age group (Fig. 4, source data in Supplementary Table S5), the results for female participants aged 60 years and above were consistent with those observed in female participants of all ages. For male participants aged 60 years and above, an increased risk for all-cause mortality was found in the unilateral near-normal, unilateral VI, and low vision groups, whereas an increased risk for CVD-related mortality was found in the unilateral near-normal group. For female participants under 60 years, an increased risk of all-cause mortality was observed only in the low vision group. Among male participants under 60 years, increased risks for all-cause and CVD-related mortality were found in the unilateral VI and the low vision groups.

Sensitivity Analyses

The results among the overall participants remained largely consistent with the main analyses after excluding those with cancer or CVD at baseline (Supplementary Table S8), excluding participants who died within the first 2 or 4 years (Supplementary Table S9), and using Fine and Gray subdistribution hazard models (Supplementary Table S10). However, the associations were slightly attenuated in some visual acuity categories.

Discussion

In this prospective cohort of 131,468 participants from the UK Biobank, we found that decreased visual acuity, including WHO-defined VI, was associated with increased risks of all-cause and CVD-related mortality. Additionally, we believe that our study was the first to report the association of increased risks for all-cause and CVD-related mortality with slightly and significantly decreased unilateral visual acuity in the WHO non-VI groups.

With the variations of visual acuity classification methods, reference groups, and study population, there were inconsistencies in findings across studies. In the UK Biobank cohort, Zhang et al. found that participants with WHO-defined VI (presenting visual acuity of the better eye worse than 0.3 LogMAR) had a 26% increased risk for all-cause mortality, combining those with mild VI and moderate to severe VI as one group and setting the participants without dual sensory impairment (VI and hearing impairment) as the reference group.[28] Supporting this, we also found that participants with WHO-defined VI exhibited a 28% increased risk for all-cause mortality setting WHO-defined non-VI group as reference. Additionally, we examined the associations between WHO-defined VI and different VI levels (mild VI and moderate to severe VI) with all-cause, CVD-related, and cancer-related mortality, and compared the results using different reference groups. For individuals within the same visual acuity category, outcomes can vary depending on the choice of reference group. In our study, when bilateral normal vision was used as the reference group, participants classified as mild VI and low vision showed higher adjusted HRs for all-cause mortality compared to when the WHO non-VI group was used as the reference. Moreover, for participants with mild VI, an increased risk for CVD-related mortality was observed only when bilateral normal vision served as the reference group, but not when the WHO non-VI group was used (see Supplementary Table S3). Of note, the impact of VI may be underestimated when the WHO non-VI group was used as the reference group because the WHO non-VI group included the bilateral normal vision, unilateral near-normal vision, bilateral near-normal vision, and unilateral VI groups. In addition, for individuals in the WHO non-VI group, the vision levels definition methods may contribute to different results. Among the young to middle-aged Korean population, Han et al. used habitual distance visual acuity in the better eye to categorize vision levels and found that individuals with mildly reduced visual acuity (between 0.5 and 0.8, decimal value) had an increased risk of all-cause mortality, but not CVD-related mortality, compared with those with normal vision (better than 0.8, decimal value).[17] In our study, which classified visual acuity category based on the combined visual acuities of both eyes, participants under 60 years with unilateral VI had increased risks of both all-cause and CVD-related mortality, whereas those with bilateral near-normal vision had an increased risk of CVD-related mortality. The method of defining vision levels based on the visual acuity of the better eye assumes that individuals with decreased unilateral visual acuity

face the same mortality risk as those with good visual acuity in both eyes, potentially overlooking the increased risk among those with unilateral visual acuity decline.

In our study, participants with decreased unilateral visual acuity (unilateral near normal and unilateral VI) and WHO-defined VI (mild VI and low vision) faced with increased risks of all-cause and CVD-related mortality. The association between VI and mortality risk was supported in other studies,[13–18] with chronic stress being a potential contributing factor. Zheng et al. suggested that increased allostatic load (AL) served as a mediator between VI and elevated all-cause and CVD-related mortality risks.[18] AL is a biomarker reflecting chronic stress and associated with an increased risk for CVD incidence.[29] Furthermore, high AL is associated with low socioeconomic status and a lack of social support.[30] Cross-sectional studies using similar visual acuity categories also showed that decreased unilateral visual acuity was associated with lower socioeconomic status, lack of support, and poorer general health.[19,20] Individuals with decreased unilateral visual acuity may experience more significant chronic stress due to these socioeconomic, mental, and physical challenges, potentially leading to an elevated AL that increases mortality risk. Additionally, lower socioeconomic status and limited support may reduce their accessibility to healthcare, further contributing to the elevated mortality risk. Unlike those with decreased unilateral visual acuity, individuals in the bilateral near-normal vision group did not exhibit lower socioeconomic status or limited support compared with those with bilateral normal vision.[19] This may partly explain the comparable risks of all-cause and CVD-related mortality observed between the two groups. On the other hand, compared with the unilateral near-normal vision (7.76%) and unilateral VI (13.18%) groups, the relatively small sample size of the bilateral near-normal vision group (0.87%) may have contributed to the lack of statistically significant associations with mortality risk. This finding warrants further investigation in other populations with different distributions of visual acuity categories.

In our study, older paticipants (aged 60 years and above) with either unilateral near-normal or unilateral VI demonstrated increased risks for all-cause, CVD-related and cancer-related mortality. Among younger participants (under 60 years), those with unilateral VI showed increased risks for all-cause and CVD-related mortality. Previous studies have suggested that limitations in daily living activities, particularly walking ability, may mediate the relationship between VI and mortality in older adults.[18,31,32] Furthermore, decreased unilateral visual acuity may impair mobility in older adults due to imbalanced visual acuity between the eyes. In younder adults, only more severe unilateral impairment may similarly hinder mobility and contribute to adverse health outcomes. However, to our knowledge, no studies have specifically examined the effects of decreased unilateral visual acuity or visual imbalance on daily living activities. In further subgroup analyses stratified by sex, unilateral VI was associated with CVD-related mortality in older female subjects but not in older male subjects. Human and animal researches suggest that reproduction is a significant stress source for female subjects, which would cause oxidative stress contributing to elevated AL.[33] The potential association between reproduction and AL, as well as the sex-specific difference in the cardiovascular biomarkers of AL might partly explain the sex difference.[30,34] The mechanisms linking decreased unilateral visual acuity to higher mortality risks are likely multifaceted, involving factors such as social status, access to healthcare, mobility limitations, and overall physical health. The UK Biobank Eye and Vision Consortium suggested these visual acuity categories as a key sensory health indicator, considering it both as a risk factor and a health outcome.[19] These complexities warrant further investigation to better understand and address these associations.

In existing studies focusing on VI, the decreased visual acuity and the comprehensive assessment of visual acuity in both eyes among individuals without VI have been neglected. Our study revealed that the classification of VI based on the visual acuity of the better eye not only overlooked the adverse effects of mildly reduced visual acuity, particularly in cases of unilateral visual acuity decline, but also underestimated the impact of VI when the WHO non-VI classification was used as the reference group. To advance ophthalmology research and policy, a broader perspective on the visual acuity category method needs to be considered. For instance, in our study, individuals with unilateral near-normal vision and 82.81% of those with unilateral VI were classified as having normal vision when visual acuity category was determined by the visual acuity of the better eye. As a result, the increased mortality risk associated with impaired visual acuity in these groups would have been overlooked. Although the distribution of visual acuity categories may vary across populations and countries, addressing the needs of previously neglected groups could deliver significant public health benefits. In clinical practice, ophthalmologists and optometrists should consider not only focusing on enhancing the vision of patients with severe VI but also addressing unilat-

eral visual acuity decline to achieve bilateral vision balance, which may contribute to reducing mortality risks. For example, in individuals with good visual acuity in one eye but uncorrected refractive error in the other, prescribing appropriate refractive correction (e.g. glasses or contact lenses) can help achieve balanced bilateral visual acuity. In older adults with bilateral cataracts, timely surgery for the fellow eye after the surgery of the first eye needs to be considered.

To our knowledge, our study novelly reported the association of the full spectrum of visual acuity categories, which was assessed using combined visual acuities of both eyes, with all-cause, CVD-related, and cancer-related mortality. With the large-scale participants, rich data sources, and long follow-up time of the UK Biobank, subgroup analysis and sensitivity analysis were conducted to verify the result's reliability with multivariable models. The comparison between the associations of the visual acuity categories and WHO VI classification with risks of all-cause, CVD-related, and cancer-related mortality was investigated.

Nevertheless, several limitations of our study warrant mention. First, because of the nonrandom recruitment and low response rate of the UK Biobank and the fact that the eye examination was performed on a subset of the population, the distribution of visual acuity categories in our study may vary among the populations in the UK and other countries.[22] However, the "healthy volunteer effect" in the UK Biobank had a limited effect on the investigation of participants with mildly reduced visual acuity because of the relatively less impacts caused by impaired vision. Second, the limited numbers of participants with moderate VI and severe VI precluded separate association analyses. These participants were considered as the low vision group, leading to masking group differences. Third, the specific eye diseases leading to decreased visual acuity were not investigated in detail. All-cause impaired habitual visual acuity was an overall visual acuity outcome of diverse eye conditions and diseases, which might have a better reflection than specific eye diseases.

In conclusion, our study found that decreased unilateral visual acuity was associated with increased risks of all-cause and CVD-related mortality among participants without VI in WHO classification in the UK Biobank. Additionally, the associations between decreased unilateral visual acuity and mortality risks varied across different sex and age subgroups. These findings highlighted the importance of adopting a broader concept of visual acuity classification method beyond VI in ophthalmology and visual sciences. Integrating strategies to achieve bilateral vision balance in clinical practice may enable ophthalmologists and optometrists to help reduce the mortality risk of individuals with decreased unilateral visual acuity.

Acknowledgments

Supported by the Ningbo Top Medical and Health Research Program (grant number 2023030716); the Medical Health Science and Technology Project of Zhejiang Province (grant number 2024XY040).

Disclosure: **C. Fu**, None; **J. Chen**, None; **Q. Gu**, None; **Y. Li**, None; **X. Li**, None; **X. Zhang**, None; **W. Chen**, None

[*] CF and JC contributed equally to this article and are to be considered co-first authors.

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